

# Project Management Charts, KPIs, and Metrics (Undergraduate & Postgraduate)

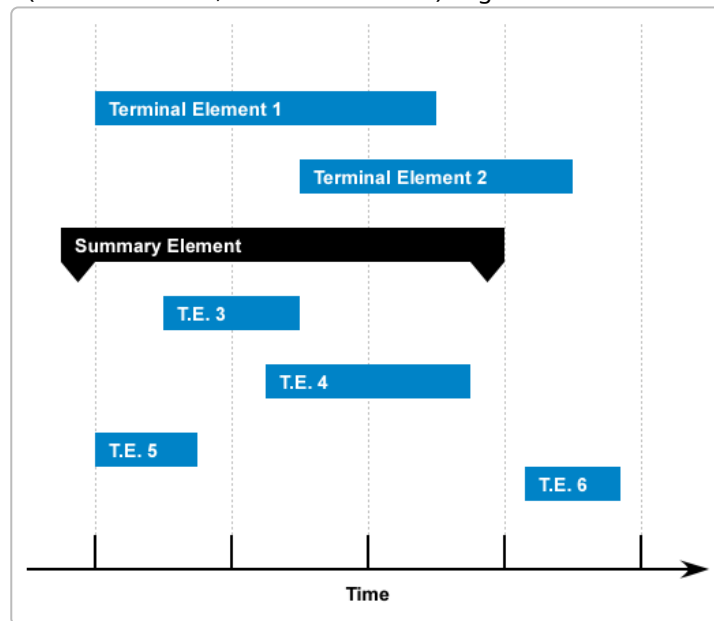
## Undergraduate Level

### Charts & Visualizations

- **Gantt Chart** – A visual timeline of project tasks, milestones, dependencies, and durations <sup>1</sup>. It's used in planning and tracking (especially in Waterfall projects) to see start/end dates and progress of each task. Required inputs include a task list with start dates, durations or end dates, and any dependencies. There is no complex math – each task is shown as a horizontal bar on a time axis. To teach/implement, students can use spreadsheet tools or MS Project: enter tasks, durations, link dependencies, then display the bar chart. *Example:*

| Task | Start Day | End Day | Duration (days) | Predecessors |
|------|-----------|---------|-----------------|--------------|
| A    | 1         | 4       | 3               | –            |
| B    | 4         | 9       | 5               | A            |
| C    | 2         | 5       | 3               | –            |
| D    | 9         | 12      | 3               | B, C         |

A mock Gantt bar chart (tasks on vertical, time on horizontal) might look like this



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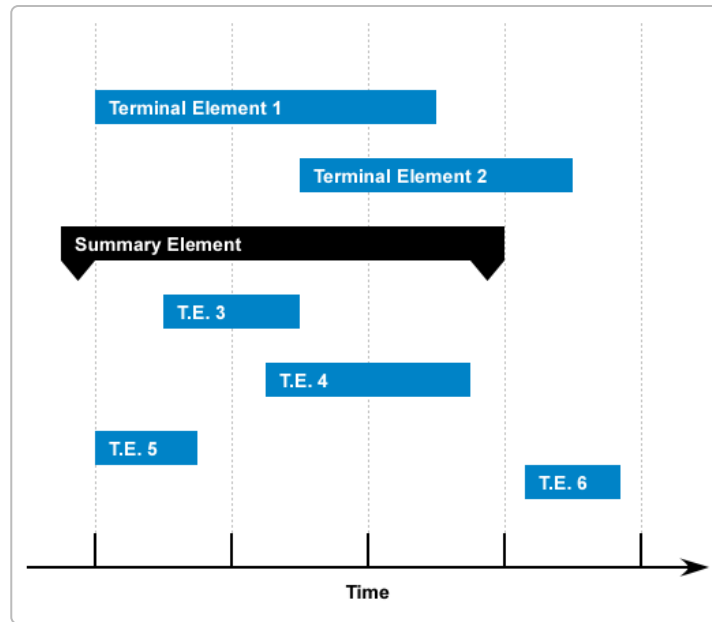


Figure: Gantt chart example (tasks vs. timeline) <sup>1</sup> .

- **Critical Path Network Diagram** – A dependency network showing all tasks and their links, used to find the “critical path” (longest-duration path) which determines project length. Input data: tasks with durations and predecessor-successor links. Math basis: forward and backward pass calculations (earliest start/finish and latest start/finish) to compute floats. In teaching, one constructs the network on paper or in software (e.g. MS Project’s network view) and computes ES, EF, LS, LF for each task; the critical path is where  $(LS-ES)=0$ . *Example:* Tasks  $A \rightarrow B \rightarrow D$  and  $A \rightarrow C \rightarrow D$ , with durations as above; compute EF/LS to find that both paths might tie or one is longest.
- **Earned Value S-Curve Chart** – A cumulative line chart plotting *Planned Value (PV)*, *Actual Cost (AC)*, and *Earned Value (EV)* over time <sup>2</sup> . Used in monitoring performance:  $PV$  = budgeted cost planned to date,  $EV$  = budgeted cost of completed work,  $AC$  = actual spend. Input: time-indexed  $PV$ ,  $EV$ ,  $AC$  values (often per period). Math: cumulative sums. Teaching: students compute  $PV$  (from schedule),  $EV$  (budget $\times\%$ complete),  $AC$  (from finance) and plot lines. The chart’s shape (an “S-curve”) reveals schedule and cost performance <sup>2</sup> .

| Week | PV (budgeted) | EV (earned) | AC (actual) |
|------|---------------|-------------|-------------|
| 1    | 10,000        | 8,000       | 9,000       |
| 2    | 25,000        | 20,000      | 22,000      |
| 3    | 45,000        | 40,000      | 44,000      |
| ...  | ...           | ...         | ...         |

The plotted example (from industry) shows  $PV$ ,  $EV$ ,  $AC$  lines over time <sup>3</sup> :

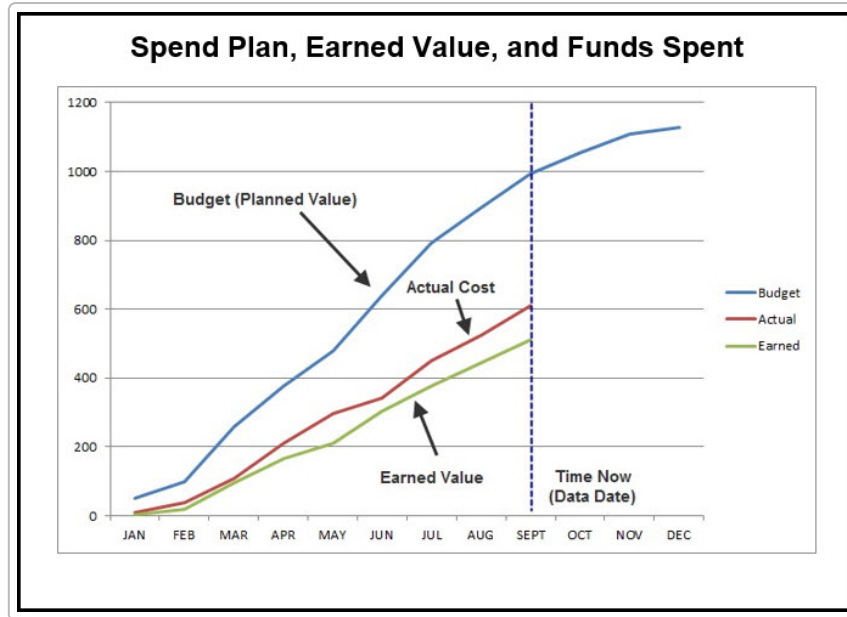
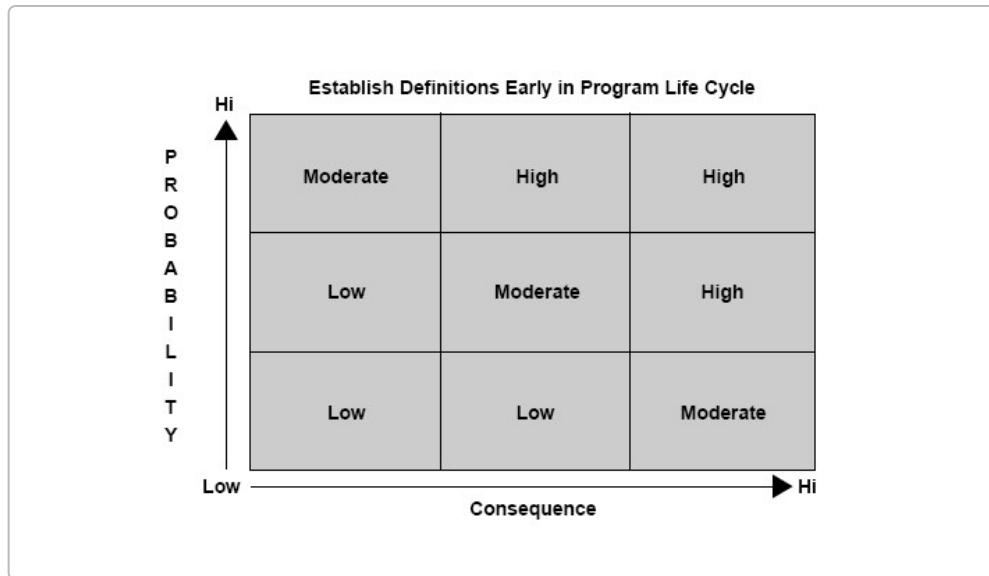


Figure: Earned-Value chart (PV = green, EV = blue, AC = red) showing performance over project life 2 .

- **Risk Impact-Probability Matrix** – A 2D grid (heat-map) categorizing risks by likelihood vs. impact. Used in risk assessment/planning: high-probability & high-impact risks are “hot” (red), low-probability/low-impact are “cool” (green). Input: list of identified risks each with an estimated probability (e.g. 0–1) and impact (e.g. \$). No heavy math – each risk is placed into a cell (e.g. 5×5 matrix). Teaching: show how to color-code and rank risks. *Example:*

| Risk               | P (0–1) | Impact (\$) |
|--------------------|---------|-------------|
| Delay in delivery  | 0.4     | 50,000      |
| Budget overrun     | 0.2     | 100,000     |
| Key staff turnover | 0.1     | 80,000      |

A risk matrix (public-domain example) illustrates colors for combinations



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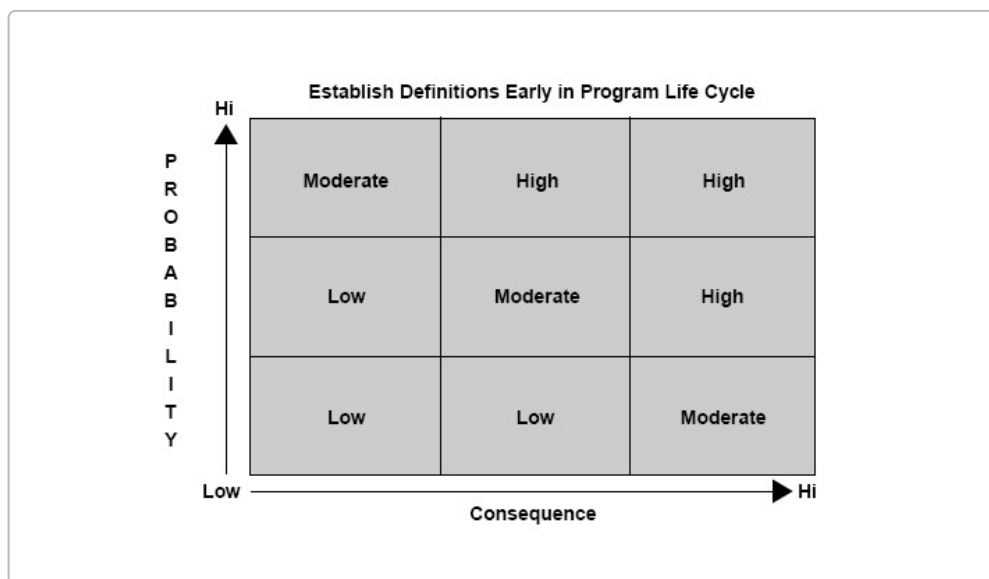


Figure: Simple risk matrix (probability vs. impact) <sup>4</sup> .

One can also compute each risk's **Risk Exposure** = *Probability* × *Impact* (expected loss) <sup>4</sup> and prioritize by value. For example, Exposure = 0.4×50k = \$20k.

## KPIs & Metrics

Project KPIs at the undergraduate level focus on basic earned-value and schedule measures:

- **Cost Variance (CV)** – Formula:  $CV = EV - AC$  <sup>5</sup> . Here EV = budgeted cost of work performed, AC = actual cost incurred. *Interpretation*: Positive CV (green) means under budget (good); negative CV (red) means over budget (bad) <sup>5</sup> . Thresholds: e.g.  $CV \geq 0$  (green), slightly negative (yellow), large

negative (red). CV is a **lagging** indicator (shows past cost performance). *Teaching:* emphasize computing EV and AC from data and that  $CV = \text{budgeted value earned} - \text{actual spend}$  <sup>5</sup>.

- **Schedule Variance (SV)** – *Formula:*  $SV = EV - PV$  <sup>6</sup>. PV = budgeted cost of work *planned* by now. Positive SV means ahead of schedule (good), negative means behind (bad) <sup>6</sup>. Use green/yellow/red by magnitude. SV is lagging (past schedule performance). *Teaching:* compare EV vs. planned PV; highlight that SV separates schedule progress from cost, using only budget values <sup>6</sup>.
- **Cost Performance Index (CPI)** – *Formula:*  $CPI = EV/AC$  <sup>5</sup>. A ratio of earned value to actual cost.  $CPI > 1.0$  means cost efficiency (under budget per work),  $<1.0$  means cost overrun <sup>5</sup>. Threshold: around 1.0 (green if  $\geq 1$ , yellow around 0.95–1.05, red if well below 1). Lagging metric. *Teaching:* show that CPI is dimensionless and can forecast cost (e.g.  $EAC = BAC/CPI$  if trend holds) <sup>5</sup>.
- **Schedule Performance Index (SPI)** – *Formula:*  $SPI = EV/PV$  <sup>6</sup>. Ratio of earned value to planned value.  $SPI > 1$  means ahead of schedule (more work done than planned),  $<1$  means lagging. Thresholds similarly around 1. Lagging indicator (measures past schedule execution) <sup>6</sup>. *Teaching:* emphasize SPI's use in predicting schedule completion (e.g. if SPI stays constant). Note: like CPI, SPI is technically retrospective ("lagging") <sup>7</sup>, but it helps signal future schedule trends.
- **Percent Complete (PC)** – *Formula:*  $PC = EV/BAC \times 100\%$  <sup>8</sup>, where BAC = total budget at completion. Indicates percent of total work (by budget) completed. Higher is better; targets (green) near 100% at end. This is a lagging status measure (reflects progress to date) <sup>8</sup>. *Teaching:* use PC to compare to planned %, or show earned-value percent complete.
- **Percent Budget Spent (PS)** – *Formula:*  $PS = AC/BAC \times 100\%$  <sup>9</sup>. Percent of budget consumed. Should be interpreted with PC. If  $PS > PC$ , you're overspending on work done. PS is lagging. *Teaching:* simple relation to AC and BAC; use alongside PC to assess efficiency.
- **Critical Ratio (CR)** – *Formula:*  $CR = CPI \times SPI$  <sup>10</sup>. Combines cost and schedule into one index. Values around 1 indicate healthy status. E.g.  $CR \approx 1$  (green), significantly  $<1$  (red). CR is lagging (draws on CPI/SPI) <sup>10</sup>. *Teaching:* highlight that  $CR \in [0.9, 1.2]$  is typical for a healthy project <sup>10</sup>.
- **Estimate at Completion (EAC)** – Forecast of total project cost. Common formula:  $EAC = AC + (BAC - EV)$  <sup>11</sup> (assuming remaining work at planned cost). Alternate:  $EAC = BAC / CPI$  (if current cost trend continues) <sup>11</sup>. Interpret:  $EAC > BAC$  means projected overrun. This is a **leading** (predictive) metric. *Teaching:* show stepwise: compute AC and EV, then plug into formula; compare EAC to BAC for variance (see VAC).
- **Variance at Completion (VAC)** – *Formula:*  $VAC = BAC - EAC$  <sup>12</sup>. Positive VAC (green) means under budget at finish; negative (red) means overrun forecast. Lagging in that it's based on current forecast. *Teaching:* emphasize VAC interprets EAC vs original plan (from formula above <sup>12</sup>).
- **To-Complete Performance Index (TCPI)** – *Formula:*  $TCPI = (BAC - EV)/(BAC - AC)$  or using EAC:  $TCPI = (BAC - EV)/(EAC - AC)$  <sup>13</sup>. Measures efficiency required on remaining work to meet BAC or current EAC.  $TCPI > 1$  (red) means remaining work must be done more efficiently than before. TCPI is leading

(required future performance). *Teaching:* use either formula; interpret  $>1$  as a signal that meeting budget needs even better cost control.

- **Project Risk Exposure** – *Formula:* for each risk,  $\$RE = p \times I\$$ , where  $p$  = probability and  $I$  = impact <sup>4</sup>; total exposure is sum over risks. Leading indicator of potential loss. Higher values require mitigation. *Teaching:* have students compute  $p \times I$  for listed risks and sum; use thresholding (e.g.  $\geq X$  flagged red).

(*Leading vs. Lagging note:* By definition, **leading** KPIs are inputs or predictors of future performance, whereas **lagging** KPIs reflect past outcomes <sup>14</sup>. In practice, EV-based KPIs (CV, SV, CPI, SPI) are **lagging** (they measure past progress), though they are often used to forecast. Forecast metrics like EAC and TCPI are **leading** (predicting future cost/schedule). Risk metrics (Exposure) are also **leading** as they quantify future risk.)

## Core Calculations

- **Critical Path (CP) Calculation:** Determine the longest path through the network. *Formula:* the project duration = sum of durations along critical path. *Steps:* list tasks with durations and dependencies; perform forward pass (compute each task's earliest start/finish) and backward pass (compute latest start/finish). Identify critical tasks (zero total float:  $LS=ES$ ). *Edge cases:* networks with multiple equal-length critical paths; cycles (invalid) or dangling tasks. *Data:* task durations, network links. *Simplification:* for students, ignore resource constraints; assume fixed durations and deterministic network. Show calculation on a small network; for example, tasks A–E, compute ES/EF and LS/LF, find that tasks A, C, D form CP of 10 days, etc. Tools like MS Project can highlight the CP.
- **Float/Slack Calculation:** *Formula:* Total Float =  $\$LF - EF\$$  (or  $\$LS - ES\$$ ). *Steps:* after CP analysis, compute each task's float. *Interpretation:* float  $>0$  means slack; float = 0 means critical. *Edge:* negative float (project schedule impossible). *Data:* from CP computation. *Teaching:* present a mini-network where one branch has slack; calculate floats manually. Emphasize floats help identify flexible tasks.
- **Earned Value (EV) Computation:** *Formula:*  $\$EV = \sum (\text{Budget}_i \times \text{\% complete}_i)\$$  over all tasks  $i$ . *Steps:* for each task, multiply its original budgeted cost by its actual completion % (physical %). Sum for all completed tasks. *Edge:* estimating %complete for non-physical deliverables; use standard techniques (e.g. 0/50/100 or detailed weightings). *Data:* WBS with task budgets and reported progress. *Simplification:* assume straightforward tasks; possibly use "budgeted hours"  $\times$  completion%.
- **Schedule and Cost Variances & Indices:** *Steps:* (1) Compute PV = budgeted cost planned by date, EV from above, AC from accounting. (2) Calculate SV = EV–PV and CV = EV–AC <sup>5</sup> <sup>6</sup>. (3) Calculate SPI = EV/PV and CPI = EV/AC <sup>5</sup> <sup>6</sup>. *Edge:* if PV or AC is zero (divide-by-zero). *Interpretation:* as above in KPIs. *Teaching:* use a sample data set; show each formula step. For example, from the EV chart table, Week 2: PV=25k, EV=20k, AC=22k  $\rightarrow$  SV=–5k (behind), SPI=0.80.
- **Estimate at Completion (EAC) Calculation:** *Steps:* choose a formula. A common one:  $\$EAC = AC + (BAC - EV)\$$  <sup>11</sup>. Alternate:  $\$EAC = BAC/CPI\$$  <sup>11</sup> (assuming CPI holds). Compute using values. *Edge:* if CPI or SPI anomalous ( $>>1$  or  $<<1$ ). *Data:* AC, EV, BAC from schedule. *Teaching:* demonstrate with the

same data above: e.g., BAC=100k, EV=40k, AC=44k →  $EAC = 44k + (100k - 40k) = 104k$  (overrun). Highlight different formulas and contexts.

- **Variance at Completion (VAC) Calculation:** *Formula:*  $VAC = BAC - EAC$  <sup>12</sup>. *Steps:* after EAC, subtract from original budget. *Interpretation:* VAC positive = underrun. *Edge:* negative means overrun (red). *Teaching:* simple subtraction after EAC; note VAC is essentially a restatement of EAC in difference form.
- **To-Complete Performance Index (TCPI):** *Formula:* e.g.  $\frac{BAC - EV}{BAC - AC}$  <sup>13</sup>. *Steps:* compute numerator (remaining work) and denominator (remaining budget), divide. *Edge:* if denominator  $\leq 0$  (no budget left). *Teaching:* highlight TCPI >1 means remaining tasks must be done more efficiently; use example numbers (BAC=100k, EV=40k, AC=50k →  $TCPI = (100 - 40) / (100 - 50) = 60 / 50 = 1.2$ , red).
- **Risk Exposure (expected loss):** *Formula:*  $RE = p \times I$  (and sum across risks) <sup>4</sup>. *Steps:* for each risk, multiply its probability by impact; sum for total exposure. *Edge:* if probabilities not independent or exposures combine (explain in advanced courses). *Data:* risk registers with  $p$  and  $I$ . *Teaching:* compute exposures for given risks; stress that  $p, I$  often from historical or expert judgment.

**Note:** In undergraduate teaching, focus on deterministic calculations. Use fixed values, simple linear relationships (no probability distributions). For example, use the standard EV formulas and network math without considering statistical variability or advanced forecasting beyond the basic formulas above.

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## Postgraduate Level

### Charts & Visualizations

- **Advanced Gantt/Timeline Views:** Similar to undergraduate Gantt, but include more details (baseline bars, percent complete shading, resource allocations). Show how tools (Primavera P6, MS Project) can overlay actual vs baseline on a Gantt. Emphasize use of custom fields (cost in hours) and linking WBS elements. *Example:* show a classroom simulation with MS Project's built-in Gantt view and "Tracking Gantt" (which shows planned vs actual bars).
- **Integrated Earned Value Dashboard:** A multi-series chart (like extended S-curve) that not only shows PV/EV/AC but also **Estimate to Complete (ETC)** and **Budget at Completion (BAC)** projections. Input historical EV/PV/AC and optional risk-adjusted forecasts. Plot trend lines extending beyond current date to predict finish cost/time. This visualization often appears in PM software dashboards. *Teaching:* simulate by extending the S-curve with forecast values using EVM formulas (e.g. extend EV using  $(CPI \times SPI)$  factor), or by linking Excel forecast functions.
- **Risk Probability Distribution Chart:** Show the probability distribution of project duration or cost (e.g. from a Monte Carlo simulation). For example, a histogram or S-curve of possible finish dates (x-axis = completion date, y-axis = probability) <sup>15</sup>. Input data: risks with distributions or three-point estimates. Basis: PERT/statistical theory. *Teaching:* demonstrate how to collect optimistic/most likely/pessimistic estimates for key tasks and run a PERT calculation: expected duration  $E = (O + 4M + P) / 6$

$6\sigma$  and standard deviation  $\sigma = (P - O)/6$ . Show resulting bell curve of project finish dates. This requires understanding of probability, so use software or spreadsheets that support distributions.

- **Resource & Cost Histograms:** Bar or line charts of resource use or cost per period (e.g. man-hours per week). Useful for analyzing resource leveling or peak usage. Input: resource assignments by time. Basis: summation of assigned hours. *Teaching:* use MS Project's resource graphs; discuss smoothing.
- **Earned Schedule Chart:** (Advanced concept) Chart comparing EV-based schedule with time, e.g. mapping "Earned Schedule" vs "Planned Schedule" to highlight schedule performance in time units rather than cost units. Input: EV and PV timeline data. *Teaching:* introduce Earned Schedule (ES) as time when PV equals EV; show chart of EV (in time) vs actual timeline.

## KPIs & Metrics

Postgraduate metrics build on undergraduate ones and add depth (forecasting, combined indices, risk, stakeholder measures):

- **Cumulative SPI/CPI Trend** – As a "leading indicator" chart, track CPI and SPI over time (e.g. week by week) to observe trends. Interpret sustained decline as warning. Thresholds: e.g. CPI or SPI trending below 1.0 (red). Leading indicator because it signals trends. *Teaching:* use historical project data to chart SPI vs time.
- **Cost and Schedule Risk Indices:** Combining EV with risk assessment. For example, **Risk-Adjusted EAC:** incorporate a contingency based on aggregate risk exposure. E.g.  $EAC_{\text{risk}} = EAC + k \times RE_{\text{total}}$ , where  $k$  depends on confidence level. *Teaching:* show how adding historical contingency or Monte Carlo will raise EAC accordingly.
- **Quality and Scope Metrics:** For completeness, mention metrics like **Defect Density** or **Change Request Rate** (though more common in agile/IT). These might be considered in advanced courses: e.g. "Number of changes per week" – leading (predicts stability) or "Completed Change Requests vs Planned" – lagging.
- **Risk Exposure (aggregate):** Extend undergraduate RE to aggregate portfolio: sum all  $p_i \times I_i$  or use probabilistic models. Postgraduate can use expected monetary value (EMV) of portfolio and set risk tolerance thresholds (e.g. green if  $<5\%$  of budget). *Teaching:* require actual probability distributions (from historical data) and expected exposure.
- **Schedule Probability of Success:** Using PERT analysis, compute probability (e.g. % on-time) given O, M, P estimates. For example, if  $TE=30$  days (mean) with  $\sigma=(P-O)/6=2$  days, compute  $P(\text{finish} \leq 32 \text{ days})$ . Leading indicator. *Teaching:* introduce normal distribution assumption for PERT and using z-scores to estimate probabilities.
- **Earned Schedule (ES) and Schedule Variance in Time:** ES = the time at which PV = EV. Then Schedule Variance (in time) = ES – current time. This is a leading schedule metric (measured in days). *Teaching:* demonstrate computing ES and interpret how many days behind/ahead (different from SV in \$.).



- **To-Complete Performance Index (TCPI)** – already introduced, but in postgraduate detail two cases: toward BAC or toward EAC. Highlight threshold: if TCPI (based on EAC) > 1 (colored red) then controlling cost will be very difficult. *Teaching:* advanced example: BAC=100k, EV=40k, AC=60k →  $TCPI(BAC) = (100-40)/(100-60) = 60/40 = 1.5$  (very high, red).  $TCPI(EAC) = (100-40)/(EAC-60)$ .
- **Monte Carlo-Based KPIs:** E.g., 50% & 90% confidence EAC and finish time, and **Probability of Meet Budget** (e.g.  $P(\text{cost} \leq \text{budget}) = x\%$ ). *Teaching:* demonstrate a simple Monte Carlo run in software and interpret the outputs.
- **Stakeholder Satisfaction / Benefit Metrics:** At postgrad level include higher-level KPIs like **Net Present Value (NPV)** or **Return on Investment (ROI)** of the project, or **Earned Business Value** (taking planned vs actual benefits). These are lagging (realized value). *Teaching:* discuss how metrics connect to business case; e.g. ROI formula for completed project. (These use financial math beyond pure project control.)

(*Leading vs. Lagging note:* At this level, emphasize which metrics are predictive. For example, **CV/SV** remain lagging, **EAC/VAC/TCPI** are predictive (leading), and **risk indices** are forward-looking. The distinction is: leading KPIs measure inputs or future-oriented variables, while lagging KPIs report past performance <sup>14</sup>.)

## Core Calculations

- **Extended Critical Path (with risk):** Use Monte Carlo to compute a probabilistic critical path. *Steps:* treat each critical task duration as a random variable (using PERT estimates or distribution). Run multiple simulations to get a distribution of project duration. *Edge:* if dependencies or limited resources, more complex. *Data:* task estimates with distributions. *Simplification:* initially use only three-point durations and assume independence. *Output:* a confidence interval for project end date. Teaching introduces tools (e.g. @Risk or Crystal Ball) or add-ins.
- **PERT Expected Time & Variance:** *Formula:*  $TE = (O + 4M + P)/6$ ,  $\sigma = (P - O)/6$  <sup>15</sup>. *Steps:* for each uncertain task, compute these. Then for critical path: expected path duration = sum  $TE$ , variance = sum  $\sigma^2$ . *Edge:* assumes beta distribution and independence. *Teaching:* show with example:  $O=15, M=24, P=30$  days →  $E=23.5$ ,  $\sigma=2.5$  (as in [43]). Compare to simple average.
- **Estimate at Completion (Advanced):** Include multiple formulas and scenario weighting. E.g. if schedule is slipping, use  $EAC = AC + \frac{BAC - EV}{CPI \times SPI}$  <sup>11</sup> (incorporates both indices). *Steps:* compute CPI and SPI, then EAC. *Teaching:* discuss when to choose each formula (e.g. if current performance is atypical, use  $AC + (BAC - EV)$ ; if structural issues, use bottom-up re-estimate).
- **Cost of Risk:** Calculate contingency reserve using expected risk. *Formula:* Total Contingency = sum( $p_i I_i$ ) + management reserve factor. *Steps:* sum exposures (as in RE) to set budget. *Edge:* risk interdependencies ignored. *Teaching:* compare with worst-case and expected-case budgeting.
- **To-Complete Performance Index (TCPI) Scenarios:** Already covered; advanced students should derive TCPI for “on budget” vs “on current forecast” goals and interpret changes.

- **Schedule Compression:** Compute impacts of crashing or fast-tracking. E.g., determine extra cost per day saved. *Steps:* analyze tasks' crash costs vs time savings. *Teaching:* simple example: task with normal cost \$10k for 5 days, crash cost \$12k for 3 days – compute cost per day saved (\$1k/day) and decision-making.
- **Monte Carlo Simulation Steps:** Lay out algorithm: identify variables (durations, costs), assign distributions, random-sample repeatedly, record outcomes. Derive metrics (mean, percentile). *Teaching:* walk through sample simulation (e.g. 1000 trials) for project finish date; plot a histogram. Emphasize importance of historical data to calibrate distributions.
- **Scaling & Complex Dependencies:** For large projects or portfolios, discuss normalization of KPIs (e.g. index values vs. absolute) and use of statistical control charts (e.g. SPI/CPI control limits) <sup>16</sup>. Also mention Earned Value at program level (aggregating multiple projects).

Throughout postgraduate instruction, stress use of real PM software (Primavera, Deltek, MS Project) for generating charts and performing analytics. Highlight how dashboards (scorecards) distill these metrics into visual trends <sup>17</sup>. Include example tables of larger data sets (e.g. multi-month EV logs) to create charts. Emphasize data requirements: many metrics need **historical performance data** (past PV, EV to compute indices) or **probabilistic estimates** (for PERT/Monte Carlo). As an example table for EV trends:

| Period | PV (cum) | EV (cum) | AC (cum) | CV  | SV   | CPI  | SPI  |
|--------|----------|----------|----------|-----|------|------|------|
| 10d    | 50k      | 45k      | 48k      | -3k | -5k  | 0.94 | 0.90 |
| 20d    | 100k     | 88k      | 95k      | -7k | -12k | 0.93 | 0.88 |
| ...    | ...      | ...      | ...      | ... | ...  | ...  | ...  |

This table feeds both charts and metrics.

Finally, underscore key concepts: **Critical Path** (determines minimum project time), **Float** (scheduling flexibility), **Earned Value** (integrates scope, time, cost), **Cost Forecasting** (EAC/VAC), and **Risk Exposure** (expected loss). Explain that advanced metrics (EAC formulas, PERT, Monte Carlo) rely on probability theory. For software demos, use built-in reports (e.g. Earned Value tables/graphs) and show how altering input data (actuals, %complete) updates all metrics.

**Sources:** Definitions and formulas from project management references <sup>18</sup> <sup>19</sup> <sup>10</sup> <sup>4</sup>, and PMI/industry guidance <sup>1</sup> <sup>7</sup> have been used to compile this instructional reference.

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<sup>1</sup> What is a Gantt Chart? Features & Benefits With Free Template | Atlassian

<https://www.atlassian.com/agile/project-management/gantt-chart>

<sup>2</sup> <sup>3</sup> Earned Value Charts Explained (With Examples) - Ten Six

<https://tensix.com/earned-value-charts-explained-with-examples/>

<sup>4</sup> What is Risk Exposure and How to Calculate It?

<https://www.metricstream.com/learn/risk-exposure.html>

5 6 11 12 19 **ProjectManagement.com - A Simple Explanation of the Earned Value Management technique**

<https://www.projectmanagement.com/blog-post/21177/A-Simple-Explanation-of-the-Earned-Value-Management-technique>

7 **pmbok - How can a given project have SPI lower than 1 and still be on schedule? - Project Management Stack Exchange**

<https://pm.stackexchange.com/questions/17880/how-can-a-given-project-have-spi-lower-than-1-and-still-be-on-schedule>

8 9 10 13 **Earned Value Management | projmgt**

<https://www.pmeducation.com/earned-value-management>

14 **ProjectManagement.com - Key Performance Indicators**

<https://www.projectmanagement.com/wikis/345150/key-performance-indicators>

15 **Three-Point Estimating and PERT Distribution (Cost & Time Estimation) - Project-Management.info**

<https://project-management.info/three-point-estimating-pert/>

16 17 **nibmehub.com**

[https://nibmehub.com/opac-service/pdf/read/Project%20management%20metrics-%20KPIs-%20and%20dashboards%20\\_%20a%20guide%20to%20measuring%20and%20monitoring%20project%20performance.pdf](https://nibmehub.com/opac-service/pdf/read/Project%20management%20metrics-%20KPIs-%20and%20dashboards%20_%20a%20guide%20to%20measuring%20and%20monitoring%20project%20performance.pdf)

18 **Schedule Variance: Definition, Formula, and How to Calculate**

<https://monday.com/blog/project-management/schedule-variance/>